Diabetes Prediction System – Evaluation Rubrics

S. No.	Evaluation Component	Criteria	Marks	Review/Deadline
1	**Problem Definition & Objectives**	Clarity of problem, motivation, objectives, scope and relevance	**5**	**Review I (40 Marks) 8/10/2026**
2	**Literature Survey & Research Gap**	Relevant papers, comparison of existing methods, identification of research gap and justification of proposed work	**8**	
3	**Dataset & Data Understanding**	Dataset selection, feature understanding, target identification, data-quality analysis and limitations	**8**	
4	**Data Preprocessing**	Missing/invalid-value handling, outlier analysis, encoding/scaling, leakage prevention and reproducible preprocessing	**5**	
5	**Exploratory Data Analysis**	Appropriate visualizations, correlation analysis, feature distributions and meaningful interpretation	**7**	
6	**Feature Engineering & Selection**	Creation/selection of meaningful features, justification and experimental comparison	**7**	
7	**Implementation**	Implementation of multiple models, correct training procedure, cross-validation and hyperparameter tuning	**10**	**Review II (40 Marks) 15/10/2026**
8	**Novelty / Proposed Framework**	Ensemble learning, calibration, uncertainty analysis, XAI, what-if analysis and integration of proposed components	**10**	
9	**Model Evaluation & Experimental Analysis**	Accuracy, precision, recall, F1, specificity, ROC-AUC/PR-AUC, calibration/Brier score, confusion matrix and statistical/experimental comparison	**8**	
10	**Explainability & Interpretation**	SHAP/local explanations, feature importance, uncertainty interpretation and responsible interpretation of results	**8**	
11	**Application / User Interface**	Functional prediction interface, input validation, visualization, model integration and usability	**4**	
12	**Documentation & Presentation**	Report quality, architecture, methodology, results, references, presentation and technical communication **0% AI content and plagiarism must be <10%**	**15**	**Review III (20 Marks) 22/10/2026**
13	**Viva & Individual Contribution**	Conceptual understanding, ability to explain code/results, troubleshooting and individual contribution	**5**	

*****